
Data Preprocessing

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Data Set and Data Object

- Data sets are made up of data objects. Most commonly a data set corresponds to the contents of a single database table, or a single statistical data matrix.
- Data object represents an entity— in a sales database, the objects may be customers, store items, and sales; in a medical database, the objects may be patients; in a university database, the objects may be students, professors, and courses. Rows of a database correspond to the data objects.

Types of Attribute

- **Nominal Attributes:** It means “relating to names.” The values of a nominal attribute are symbols or names of things.
 - Each value represents some kind of category, code, or state. The values do not have any meaningful order known as enumerations. *Hair_color* and *Marital_status* are two nominal attributes of *Person* objects.
- **Binary Attributes:** Binary attribute is a nominal attribute with only two categories or states: 0 or 1, where 0 typically means that the attribute is absent, and 1 means that it is present.
 - It referred to as *Boolean* if the two states correspond to true and false. A binary attribute can be symmetric if both of its states are equally valuable and carry the same weight other wise it will be asymmetric. As for example, attribute *smoker* describes a *Patient* object.

Types of Attribute

- **Ordinal Attributes:** It is an attribute with possible values that have a meaningful order or ranking among them, but the magnitude between successive values is not known.
 - Example of ordinal attributes includes grade (e.g., A+, A, A-, B+, and so on). Customer satisfaction had the following ordinal categories: 0: very dissatisfied, 1: somewhat dissatisfied, 2: neutral, 3: satisfied, and 4: very satisfied.
- **Numeric Attributes:** A numeric attribute is quantitative; that is, it is a measurable quantity, represented in integer or real values. Numeric attributes can be of two types:
 - **Interval-scaled:** measured on a scale of equal-size units
 - **Ratio-scaled:** measured as multiple (or ratio) of another value

Types of Attribute

- **Quantitative data** deals with numbers and things you can measure objectively: dimensions such as height, width, and length, Temperature and humidity, Prices, Area and Volume.
- **Qualitative data** deals with characteristics and descriptors that can't be easily measured, but can be observed subjectively—such as smells, tastes, textures, attractiveness, and color.

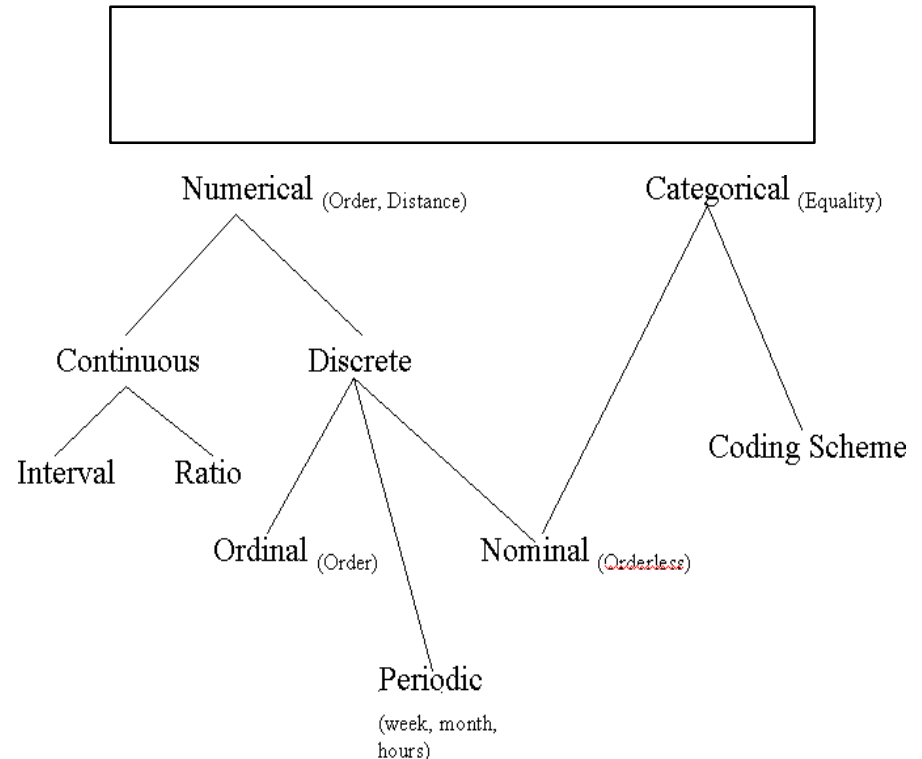
Nominal, binary, and ordinal attributes are qualitative. That is, they describe a feature of an object without giving an actual size or quantity.

Types of Attribute

- **Discrete attribute** has a *finite* or *countably infinite* set of values, which may or may not be represented as integers. It may have numeric values, such as 0 and 1 for binary attributes or, the values 0 to 110 for the attribute age.
 - An attribute is *countably infinite* if the set of possible values is infinite but the values can be put in a one-to-one correspondence with natural numbers. Zip codes are another example.
- **Continuous attribute** is described as an unbroken set of observations; that can be measured on a scale. It can take any numeric value, within a finite or infinite range of possible value.
 - Statistically, range refers to the difference between highest and lowest observation. The continuous data can be broken down into fractions and decimal, i.e., it can be meaningfully subdivided into smaller parts according to the measurement precision. Continuous attributes are typically represented as floating-point variables.

Data Types and Forms

- Attribute-value data:
- Data types
 - numeric, categorical ([see the hierarchy for its relationship](#))
 - static, dynamic (temporal)
- Other kinds of data
 - distributed data
 - text, Web, meta data
 - images, audio/video



Why Data Preprocessing?

- Data in the real world is dirty
 - **incomplete**: missing attribute values, lack of certain attributes of interest, or containing only aggregate data
 - e.g., occupation=""
 - **noisy**: containing errors or outliers
 - e.g., Salary="-10"
 - **inconsistent**: containing discrepancies in codes or names
 - e.g., Age="42" Birthday="03/07/1997"
 - e.g., Was rating "1,2,3", now rating "A, B, C"
 - e.g., discrepancy between duplicate records

Why Is Data Preprocessing Important?

- No quality data, no quality mining results!
 - Quality decisions must be based on quality data
 - e.g., duplicate or missing data may cause incorrect or even misleading statistics.
- Data preparation, cleaning, and transformation comprises the majority of the work in a data mining application (90%).

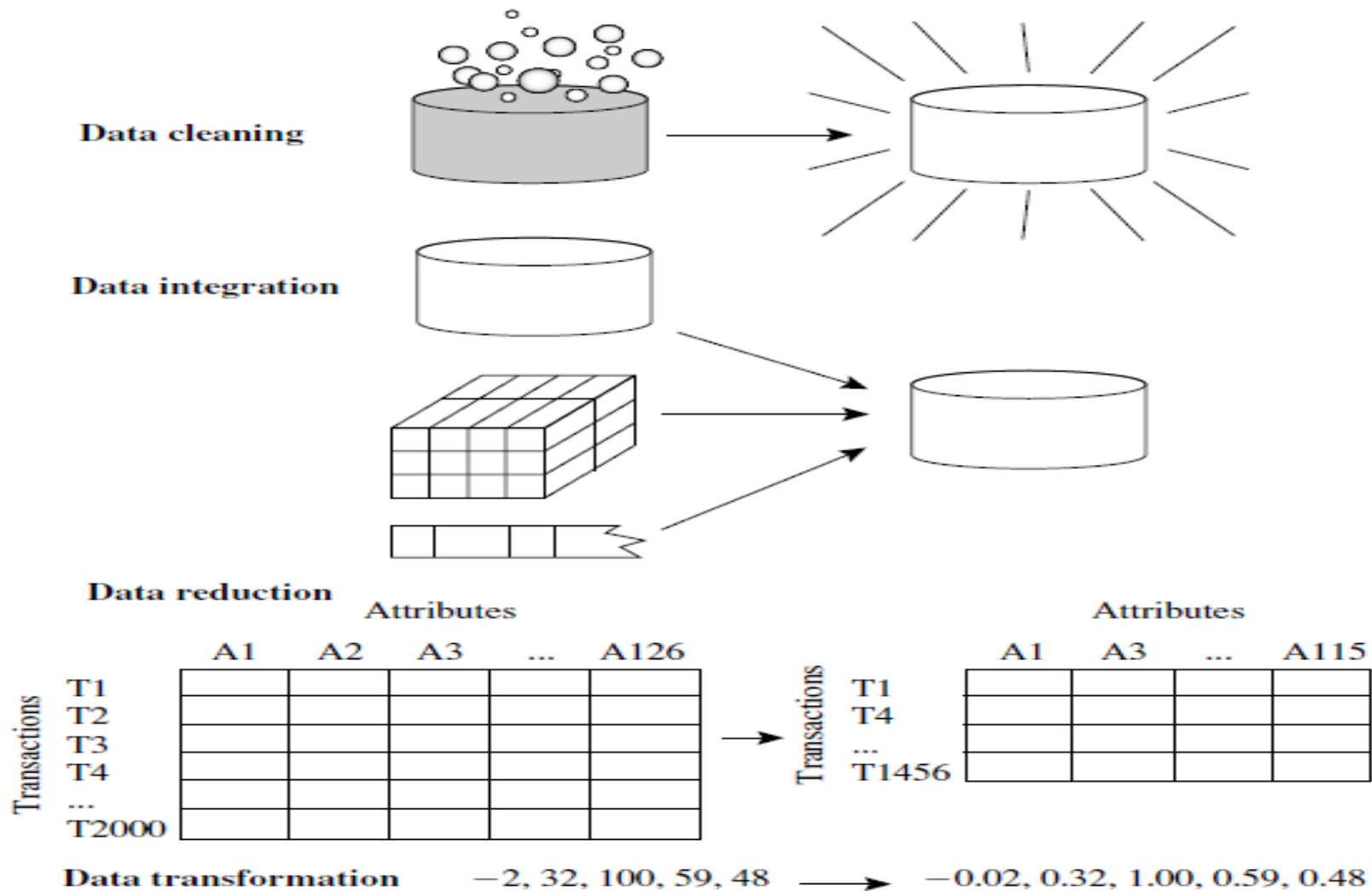
Multi-Dimensional Measure of Data Quality

- A well-accepted multi-dimensional view:
 - Accuracy
 - Completeness
 - Consistency
 - Timeliness
 - Believability
 - Value added
 - Interpretability
 - Accessibility

Major Tasks in Data Preprocessing

- Data cleaning
 - Fill in missing values, smooth noisy data, identify or remove outliers and noisy data, and resolve inconsistencies
- Data integration
 - Integration of multiple databases, or files
- Data transformation
 - Normalization and aggregation
- Data reduction
 - Obtains reduced representation in volume but produces the same or similar analytical results
- Data discretization (for numerical data)

Major Tasks in Data Preprocessing



Data Preprocessing

- Why preprocess the data?
- Data cleaning
- Data integration and transformation
- Data reduction
- Discretization
- Summary

Data Cleaning

- Importance
 - “Data cleaning is the number one problem in data warehousing”
- Data cleaning tasks
 - Fill in missing values
 - Identify outliers and smooth out noisy data
 - Correct inconsistent data
 - Resolve redundancy caused by data integration

Missing Data

- Data is not always available
 - E.g., many tuples have no recorded values for several attributes, such as customer income in sales data
- Missing data may be due to
 - equipment malfunction
 - inconsistent with other recorded data and thus deleted
 - data not entered due to misunderstanding
 - certain data may not be considered important at the time of entry
 - not register history or changes of the data

How to Handle Missing Data?

- Ignore the tuple
- Fill in missing values manually: tedious + infeasible?
- Fill in it automatically with
 - a global constant : e.g., “unknown”, a new class?!
 - the attribute mean
 - the most probable value: inference-based such as Bayesian formula, decision tree, or EM algorithm

Noisy Data

- Noise: random error or variance in a measured variable.
- Incorrect attribute values may due to
 - faulty data collection instruments
 - data entry problems
 - data transmission problems
 - etc
- Other data problems which requires data cleaning
 - duplicate records, incomplete data, inconsistent data

How to Handle Noisy Data?

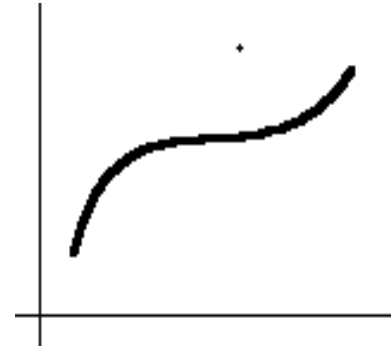
- Binning method:
 - first sort data and partition into (equi-depth) bins
 - then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
- Clustering
 - detect and remove outliers
- Combined computer and human inspection
 - detect suspicious values and check by human (e.g., deal with possible outliers)

Binning Methods for Data Smoothing

- Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34
- Partition into (equi-depth) bins:
 - Bin 1: 4, 8, 9, 15
 - Bin 2: 21, 21, 24, 25
 - Bin 3: 26, 28, 29, 34
- Smoothing by bin means:
 - Bin 1: 9, 9, 9, 9
 - Bin 2: 23, 23, 23, 23
 - Bin 3: 29, 29, 29, 29
- Smoothing by bin boundaries:
 - Bin 1: 4, 4, 4, 15
 - Bin 2: 21, 21, 25, 25
 - Bin 3: 26, 26, 26, 34

Outlier Removal

- Data points inconsistent with the majority of data
- Different outliers
 - Valid: CEO's salary,
 - Noisy: One's age = 200, widely deviated points
- Removal methods
 - Clustering
 - Curve-fitting
 - Hypothesis-testing with a given model



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Data Integration

- Data integration:
 - combines data from multiple sources
- Schema integration
 - integrate metadata from different sources
 - Entity identification problem: identify real world entities from multiple data sources, e.g., A.cust-id $\equiv$ B.cust-#
- Detecting and resolving data value conflicts
 - for the same real world entity, attribute values from different sources are different, e.g., different scales, metric vs. British units
- Removing duplicates and redundant data

Data Integration

- Entity Identification Problem:
 - Number of issues to consider during data integration such as **Schema integration** and **object matching** can be tricky.
 - How can equivalent real-world entities from multiple data sources be matched up?
- Redundancy and Correlation Analysis:
 - For **nominal data**, we use the **chi-square test**. For **numeric attributes**, we can use the **correlation coefficient** and **covariance**, both of which access how one attribute's values vary from those of another.
- Tuple Duplication- removing duplicate tuples.
- Data Value Conflict Detection and Resolution

Data Transformation

- Smoothing: remove noise from data
- Normalization: scaled to fall within a small, specified range
- Attribute/feature construction
 - New attributes constructed from the given ones
- Aggregation: summarization
- Generalization: concept hierarchy climbing

Data Transformation: Normalization

- min-max normalization

$$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

- z-score normalization

$$v' = \frac{v - \text{mean}_A}{\text{stand_dev}_A}$$

- normalization by decimal scaling

$$v' = \frac{v}{10^j} \quad \text{Where } j \text{ is the smallest integer such that } \text{Max}(|v'|) < 1$$

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Data Reduction Strategies

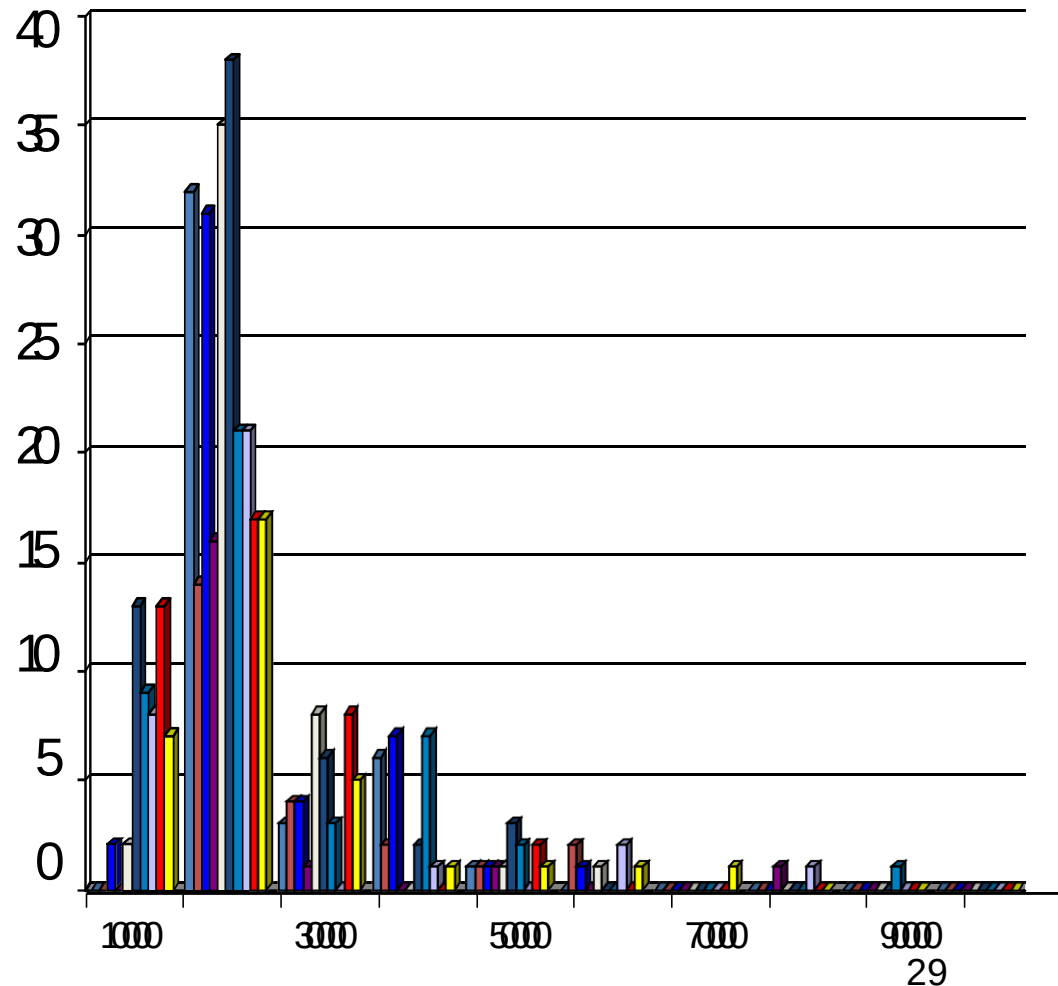
- Data is too big to work with
- Data reduction
 - Obtain a reduced representation of the data set that is much smaller in volume but yet produce the same (or almost the same) analytical results
- Data reduction strategies
 - Dimensionality reduction — remove unimportant attributes
 - Aggregation and clustering
 - Sampling

Dimensionality Reduction

- Feature selection (i.e., attribute subset selection):
 - Select a minimum set of attributes (features) that is sufficient for the data mining task.
- Heuristic methods (due to exponential # of choices):
 - step-wise forward selection
 - step-wise backward elimination
 - combining forward selection and backward elimination
 - etc

Histograms

- A popular data reduction technique
- Divide data into buckets and store average (sum) for each bucket



Clustering

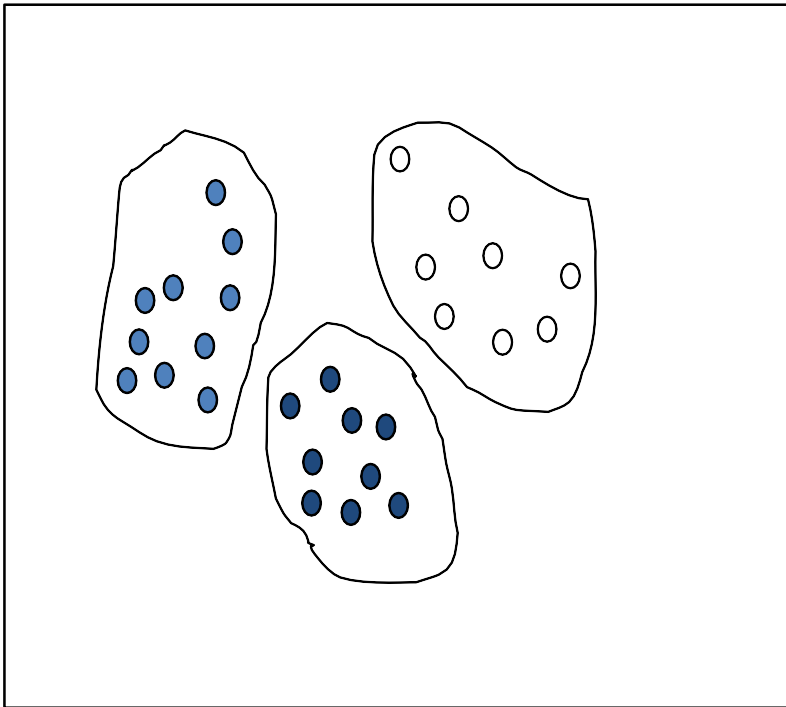
- Partition data set into clusters, and one can store cluster representation only
- Can be very effective if data is clustered but not if data is “smeared”
- There are many choices of clustering definitions and clustering algorithms. We will discuss them later.

Sampling

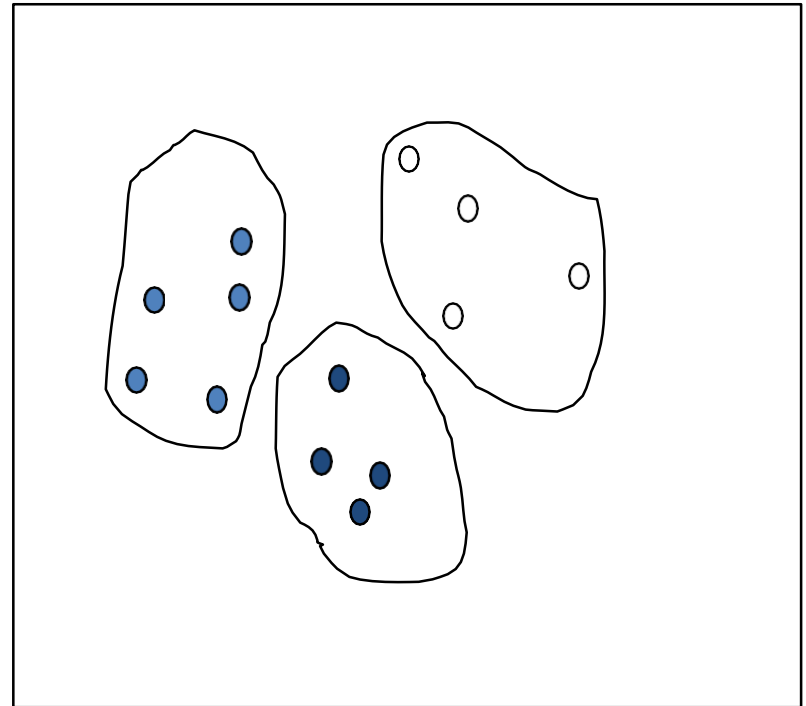
- Choose a **representative** subset of the data
 - Simple random sampling may have poor performance in the presence of skew.
- Develop adaptive sampling methods
 - Stratified sampling:
 - Approximate the percentage of each class (or subpopulation of interest) in the overall database
 - Used in conjunction with skewed data

Sampling

Raw Data



Cluster/Stratified Sample



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Discretization

- Three types of attributes:
 - Nominal — values from an unordered set
 - Ordinal — values from an ordered set
 - Continuous — real numbers
- Discretization:
 - divide the range of a continuous attribute into intervals because some data mining algorithms only accept categorical attributes.
- Some techniques:
 - Binning methods – equal-width, equal-frequency
 - Entropy-based methods

Discretization and Concept Hierarchy

- Discretization
 - reduce the number of values for a given continuous attribute by dividing the range of the attribute into intervals. Interval labels can then be used to replace actual data values
- Concept hierarchies
 - reduce the data by collecting and replacing low level concepts (such as numeric values for the attribute age) by higher level concepts (such as young, middle-aged, or senior)

Binning

- ❑ Attribute values (for one attribute e.g., age):
– 0, 4, 12, 16, 16, 18, 24, 26, 28
- ❑ Equi-width binning – for bin width of e.g., 10:
 - Bin 1: 0, 4 [-,10) bin
 - Bin 2: 12, 16, 16, 18 [10,20) bin
 - Bin 3: 24, 26, 28 [20,+) bin
 - (–) denote negative infinity, (+) positive infinity
- ❑ Equi-frequency binning – for bin density of e.g., 3:
 - Bin 1: 0, 4, 12 [-, 14) bin
 - Bin 2: 16, 16, 18 [14, 21) bin
 - Bin 3: 24, 26, 28 [21,+] bin

Entropy-based (1)

- Given attribute-value/class pairs:
 - (0,P), (4,P), (12,P), (16,N), (16,N), (18,P), (24,N), (26,N), (28,N)
- Entropy-based binning via binarization:
 - Intuitively, find best split so that the bins are as pure as possible
 - Formally characterized by maximal information gain.
- Let S denote the above 9 pairs, $p=4/9$ be fraction of P pairs, and $n=5/9$ be fraction of N pairs.
- $\text{Entropy}(S) = -p \log p - n \log n$.
 - Smaller entropy – set is relatively pure; smallest is 0.
 - Large entropy – set is mixed. Largest is 1.

Entropy-based (2)

- Let v be a possible split. Then S is divided into two sets:
 - $S1$: value $\leq v$ and $S2$: value $> v$
- Information of the split:
 - $I(S1, S2) = (|S1|/|S|) \text{Entropy}(S1) + (|S2|/|S|) \text{Entropy}(S2)$
- Information gain of the split:
 - $\text{Gain}(v, S) = \text{Entropy}(S) - I(S1, S2)$
- ❑ **Goal:** split with maximal information gain.
- Possible splits: mid points b/w any two consecutive values.
- For $v=14$, $I(S1, S2) = 0 + 6/9 * \text{Entropy}(S2) = 6/9 * 0.65 = 0.433$
 - $\text{Gain}(14, S) = \text{Entropy}(S) - 0.433$
 - maximum *Gain* means minimum *I*.
- The best split is found after examining all possible splits.

Data Transformation and Data Discretization

Concept of Hierarchy Generation:

- Hierarchy can be defined by specifying the total ordering among these attributes at the schema level such as:

street < city < province_or_state < country.



Automatic generation of a schema concept hierarchy
based on the number of distinct attribute values.

Summary

- Data preparation is a big issue for data mining
- Data preparation includes
 - Data cleaning and data integration
 - Data reduction and feature selection
 - Discretization
- Many methods have been proposed but still an active area of research